

Conversational SQL Assistant Powered by Local LLMs

QueryCrafter

Bridging the Gap Between Natural Language and Relational Databases

LANGUAGE CONVERSION

Natural → SQL

INFRASTRUCTURE

Local Ollama AI

EXECUTION MODE

100% Secure & Offline

Query Generation Lifecycle





Engineered for Speed & Safety



DDL Ingestion

Robust DDL Schema Mapping via a dedicated control panel, allowing users to input **CREATE TABLE** statements to give the AI precise domain context.



Configurable API

Configurable API Connections support various HTTP methods, custom headers, and authentication strategies for seamless integration.



Local Engine

An adaptive local results engine provides **Dynamic Data Rendering**, visualizing outputs as raw JSON payloads or structured tabular data.



Session Isolation

Instantly reset conversations, spin up distinct analytic branches, or swap out mapping structures smoothly while protecting sensitive data.



Under the Hood Stack



Frontend Layer

The architecture relies on a modern, responsive Frontend Layer built with React, utilizing hooks-driven session states and custom data table parsers.



AI Engine (Local Ollama)

The core AI Engine utilizes local LLM instances orchestrated securely via Ollama APIs, guaranteeing data privacy.



Backend Ingestion

Supporting this is an extensible Backend Layer designed for flexible JSON and CSV data ingestion, complete with automated column and row parsing to handle diverse data structures efficiently.

Driving Direct Business Impact

90%

REDUCED WAITING TIME

QueryCrafter drives immense productivity gains by eliminating repetitive SQL writing for engineering teams, freeing them for high-value tasks.

Democratizing Data Securely

It safely empowers data democratization, opening up direct database exploration to non-technical business stakeholders without compromising security.



Zero

Cloud Dependency
Costs



Roadmap

The future roadmap includes expanding capabilities to support complex multi-database joins, visual schema graphing, and automated query benchmarking.

QueryCranter: Conversational SQL Assistant UI Presentation

 **QueryCrafter**

Conversational SQL Assistant powered by Ollama

Database Schema (DDL)

```
ALTER TABLE `cart`  
ADD CONSTRAINT  
--  
UNIQUE (name, description, price, category)  
--
```

Domain

General

Loaded

New Session

Schema Loaded

API Connection Configurations

test
http://localhost:11434/api/chat
POST

New API Config

Conversation

find all the food items

Generated SQL

```
SELECT name, description, price, category  
FROM menu
```

Preview Result

TableJSON

category	description	name	price
Samosa	Spicy potato fritter	Samosa	50.00
Vada Pav	Potato burger	Vada Pav	40.00
Paneer Tikka	Grilled cottage cheese	Paneer Tikka	250.00
Paneer Tikka	Spicy fried chicken	Chicken 65	220.00
Chicken 65	Chicken 65		220.00

Ask your SQL question...